

1 METHOD SELECTION

DECISION

$h = 25 \text{ ft} \leq 60 \text{ ft}$ Part 4 (Eq. 30.6-1) does NOT apply
 – Part 4 scope is $60 \text{ ft} < h \leq 160 \text{ ft}$
 governing Part 6, §30.8, Eq. (30.8-1)
 note §30.9 is roof overhangs, not parapets

ASSUMPTIONS

roof flat, $\theta \leq 7^\circ$ (Fig. 30.3-2A)
 enclosure enclosed building
 parapet solid knee wall; no envelope $\Rightarrow (GC_{pi}) = 0$
 K_{zt}, K_e 1.0

2 INPUTS

V 153 mph
 exposure D
 h 25 ft
 h_p 3 ft
 $h + h_p$ 28 ft (top of parapet)
 studs 2×6 knee studs at 16 in. o.c.
 span 3.29 ft (3'-3.5" cantilever above deck)
 A $4.4 \text{ ft}^2 = 3.29 \times 1.33$

3 VELOCITY PRESSURE

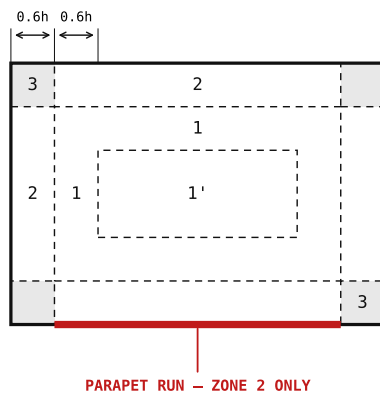
$$q_p = 0.00256 K_z K_{zt} K_d K_e V^2 \quad (\text{ASCE 7-16 Eq. 26.10-1})$$

q_p velocity pressure at top of parapet, psf
 K_z 1.144 at $z = 28 \text{ ft}$, Exp. D (Table 26.10-1, interp. 25-30 ft)
 K_d 0.85, C&C (Table 26.6-1)
 K_e 1.00 (Table 26.9-1)

$$q_p = 0.00256(1.144)(1.0)(0.85)(1.0)(153)^2 = 58.3 \text{ psf}$$

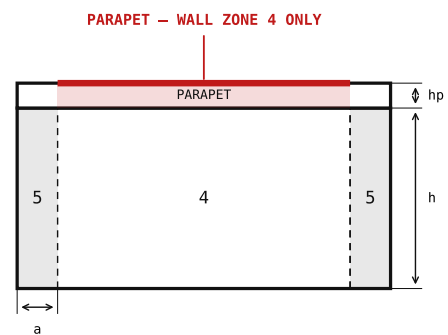
4 EXTERNAL PRESSURE COEFFICIENTS

ROOF PLAN – ASCE 7-16 FIG. 30.3-2A



$$0.6h = 0.6(25) = 15 \text{ ft} \quad \text{zone 3} = \text{corner, not applicable here}$$

WALL ELEVATION – ASCE 7-16 FIG. 30.3-1



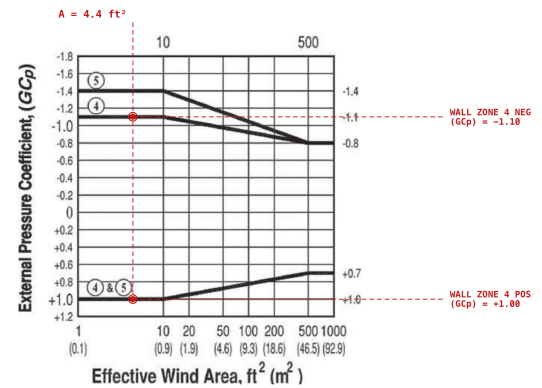
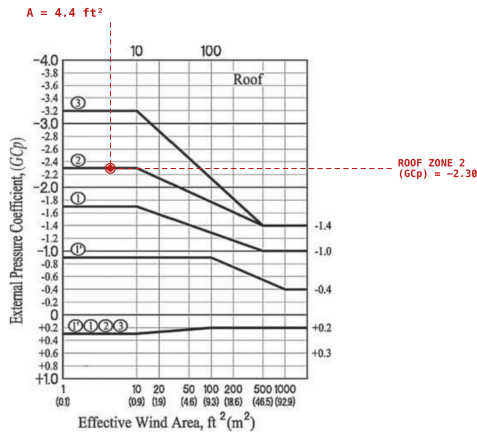
$$h = 25 \text{ ft} \quad h_p = 3 \text{ ft} \quad a = \min(0.1 \times \text{least dim}, 0.4h) \geq 3 \text{ ft}$$

WHERE: Figs. 30.3-1, 30.3-2A. $A = 4.4 \text{ ft}^2 \leq 10 \text{ ft}^2$: plateau values, no interpolation.

SURFACE	ZONE	(GC_p)
wall, positive (Fig. 30.3-1)	4	+1.00
wall, negative (Fig. 30.3-1)	4	-1.10
roof, negative (Fig. 30.3-2A)	2	-2.30

EFFECTIVE WIND AREA

element 2x6 knee stud, vertical cantilever
 A span $\times$ trib = $3.29 \times 1.33 = 4.4 \text{ ft}^2$
 \$26.2 width $1.33 \text{ ft} \geq \text{span}/3 = 1.10 \text{ ft}$
 - $A \leq 10 \text{ ft}^2 \Rightarrow$ flat of curve, most severe (GC_p)



ASCE 7-16 Fig. 30.3-2A (roof, $\theta \leq 7^\circ$) and Fig. 30.3-1 (walls); read at $A = 4.4 \text{ ft}^2$.

LIMITS

zone scope parapet run is interior edge only; no corner
 - Zone 3 (roof) and Zone 5 (wall) do not apply
 Fig. 30.3-1 Note 5 wall (GC_p) may be cut 10% when $\theta \leq 10^\circ$; not taken

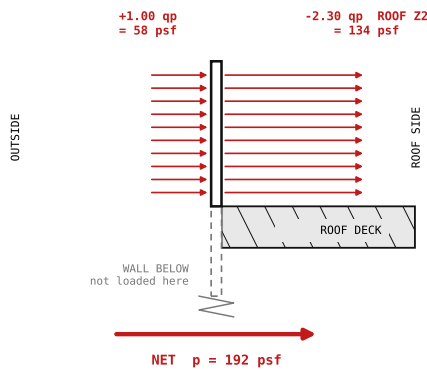
5 DESIGN PRESSURE

$$p = q_p [(GC_p) - (GC_{pi})] \quad (\text{ASCE 7-16 Eq. 30.8-1})$$

WHERE Fig. 30.8-1, two load cases per §30.8
 (GC_{pi}) per porosity of the parapet envelope; = 0 here
 Case A windward parapet: +wall front, -roof Zone 2 back
 Case B leeward parapet: +wall back, -wall Zone 4 front

CASE A – WINDWARD PARAPET

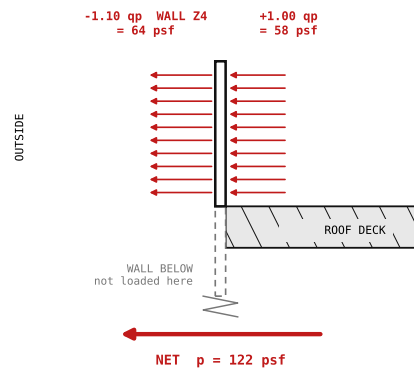
front +1.00 wall | back -2.30 roof Z2 | cantilevers off deck
 WIND $\rightarrow$



GOVERNS both faces drive the same way

CASE B – LEEWARD PARAPET

front +1.00 wall | back -1.10 wall Z4 | cantilevers off deck
 $\leftarrow$ WIND



REVERSAL design the stem for both directions

CASE	SURFACES	NET (GC_p)	p (psf)
A	wall +1.00 / roof Z2 -2.30	3.30	192
B	wall +1.00 / wall Z4 -1.10	2.10	122

6 GOVERNING

Case A $p = 192 \text{ psf}$ (windward parapet, roof Zone 2)
 direction acts horizontally, normal to parapet face